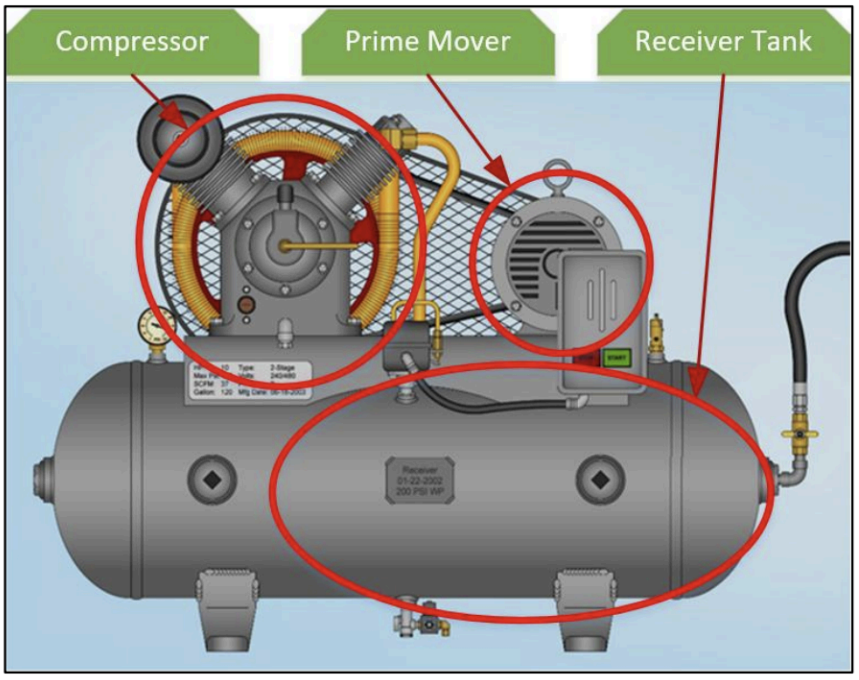
Pneumatic Control Systems

Pneumatic systems are systems powered by compressed air to do work, producing linear and rotary motion.

Pneumatic Power Supply

To supply compressed air for these systems, we use a pneumatic power supply system to provide compressed air, and in the case of this module of 6 bar/ 0.6MPa pressure.

Generally, pneumatic power supply systems consists of three parts:



The prime mover is just an electric motor that couples with the air compressor and drives the motion of the compressor, while the compressor takes in air from the atmosphere and compresses it to a higher pressure. The pressurized air is then cooled or dried before transported into the receiver tank, which stores and provides a steady supply of air, with a safety valve too to

Air Service Unit

Before being used, the compressed air must be conditioned before being used at the service point. Conditioning the air involves filtering contaminants

and water vapour, regulating the air pressure level and lubricating the air. Hence, this system is also called the FRL unit.

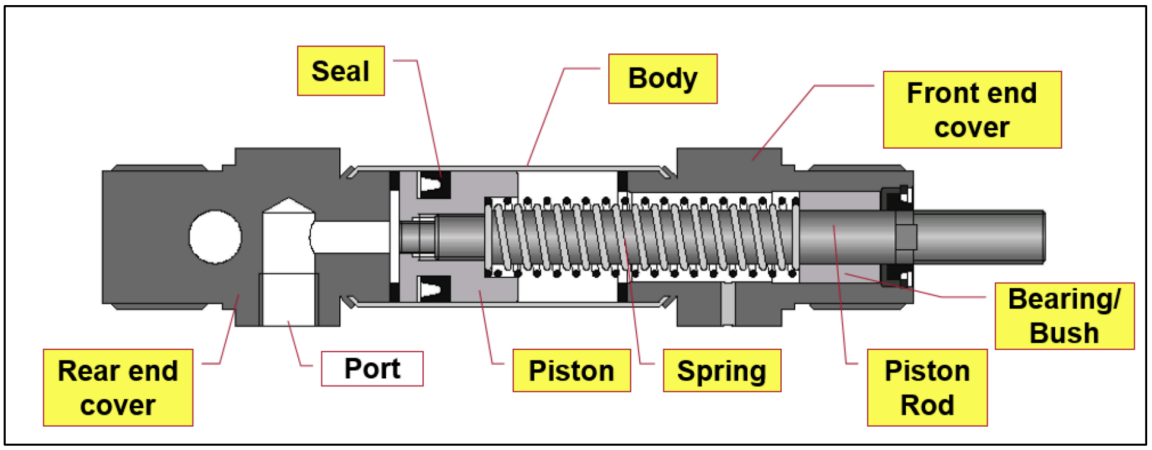
The air filter flows through an inlet that causes it to swirl, allowing centrifugal forces to push out the heavier contaminants and water particles from the air, before the air passes through a paper filter to filter out smaller particles.

Afterwards, the compressed air is regulated to the same safe pressure via the pressure regulator, and the lubricator provides a fine mist of oil, reducing friction and corrosion.

Pneumatic Cylinders

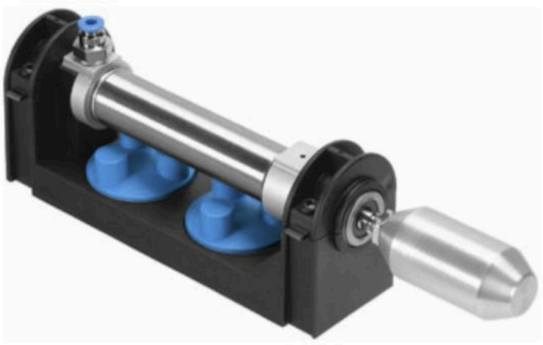
The most common type of pneumatic actuator is the pneumatic cylinder, which involves using compressed air to exert a force on a piston to move it along its cylinder. Typically, there are two kinds of pneumatic cylinders.

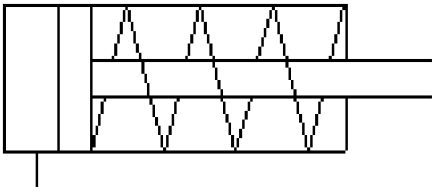
Single acting cylinders only have one port that accepts compressed air, and typically have a spring to resist the force due to the compressed air.



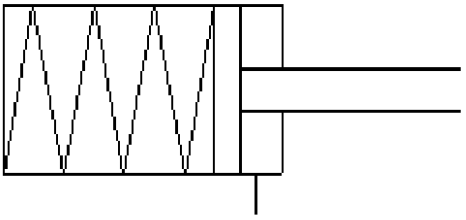
In the case of this pneumatic cylinder, the compressed air causes the spring to compress and pushes the piston and piston rod outward, hence this type of single acting cylinder is called a push-type single acting piston cylinder. Note that the compressed air has to work against the spring force and the friction to do work.

We represent push-type single acting pistons with the following ISO symbol:



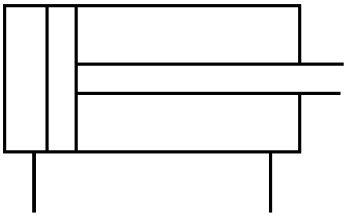


The spring may also be placed on the opposite side of the piston, to form a pull type single acting cylinder:



Work is only performed in the return stroke for this cylinder.

Double acting cylinders have two ports and no spring, and requires compressed air to be supplied to one end and exhausted from the other for work to be done. Hence, work is done on both strokes of the cylinder.



Since the airflow can be controlled, the speed of each stroke can be controlled too.

Pneumatic Cylinder Sizing

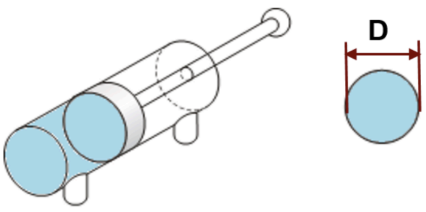
The size of a cylinder & its dimensions depends on the applications and the force needed.

Theoretically, the force by a double acting cylinder is

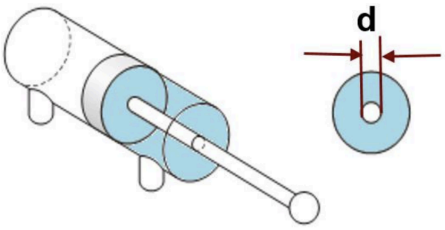
$$F_t = PA$$

where A is in mm^2 and P is in N/mm^2 or MPa.

The area depends on the type of stroke:



Forward Stroke



Return Stroke

where

$$A_{forward} = \frac{\pi D^2}{4}$$
$$A_{return} = \frac{\pi (D-d)^2}{4}$$

However, some force is consumed by friction, hence the effective force is

$$F_e = F_t - F_f$$

but since we usually assume that frictional force is 0.1 of the theoretical force,

$$F_e = 0.9 F_t$$

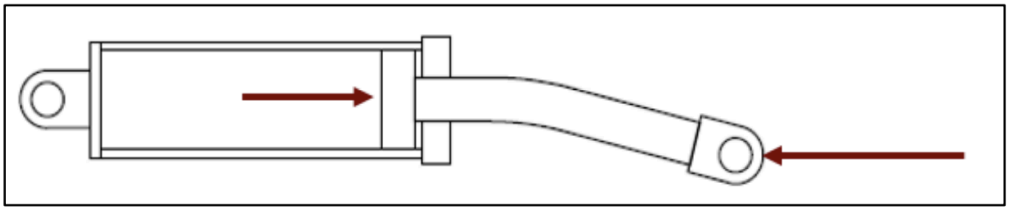
Pneumatic Cylinder Speed

The speed of the piston is mainly dependent on the air flow rate, cylinder size, and the load to force ratio (or the load ratio). Other geometrical factors affect the speed, however the main method to control the speed is by limiting the air flow into the inlets of the cylinder.

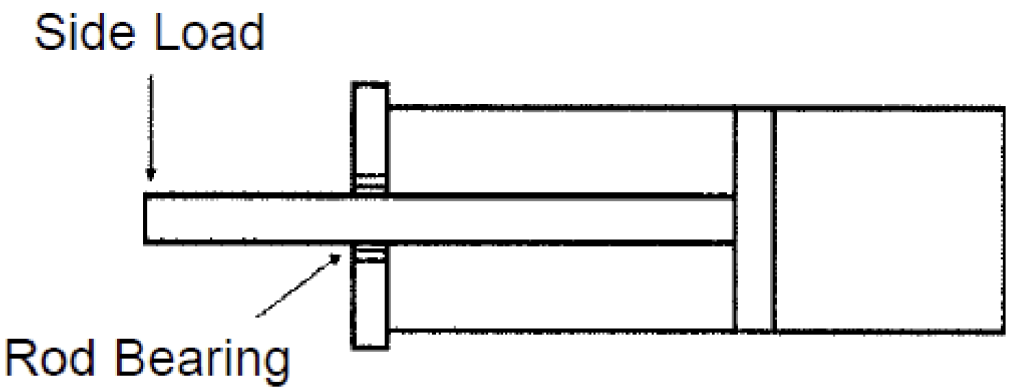
Pneumatic Cylinder Failure

There are two possible ways for a piston cylinder to fail.

The piston rod must be large enough in relation to the load column, otherwise the forces are not aligned and cause the piston rod to buckle.



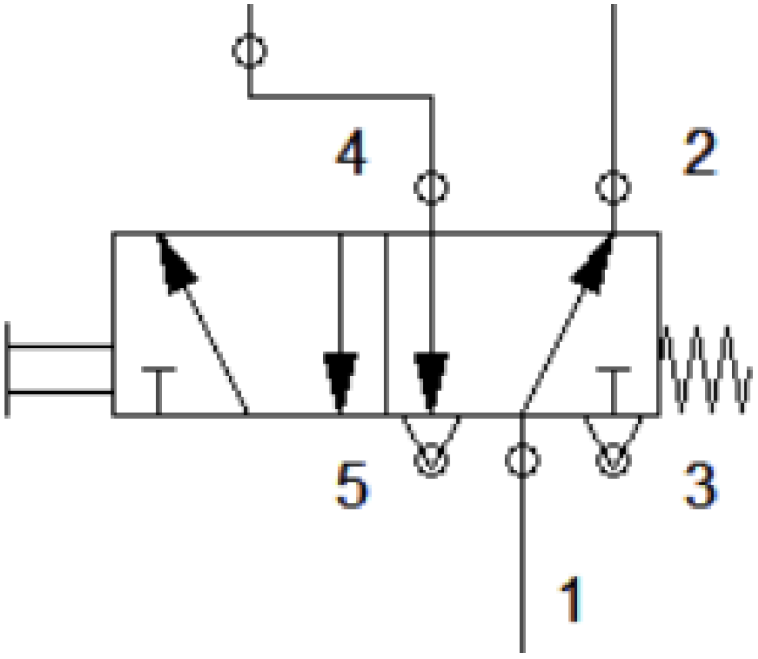
More commonly, side loading is the more frequent cause of failure. This may be due to misalignments with the load when the piston cylinder is mounted. Especially when the piston is extended, extensive leverage/radial loads are developed which reduce the life of the piston and rod bushing.ss



Pneumatic Valves

To control the flow of air for pneumatic cylinders, we use pneumatic valves.

Shown is a typical valve graphical symbol:



We describe directional valves by the

Number of ports (in this case 5)

Number of positions (represented as boxes, in this case 2)

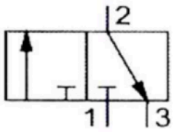
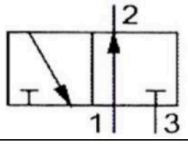
Method of Actuation (manual, mechanical, electrical, electromechanical, pnuematic)

Methods of Return actuation (spring, pneumatic, etc)

Any other special features

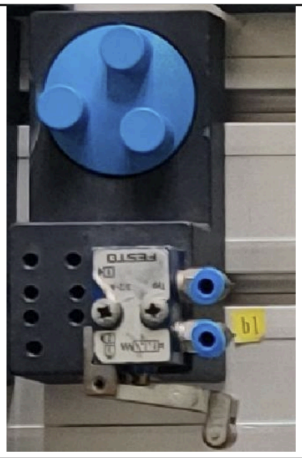
In this case, this is a 5/2-way manual directional control valve.

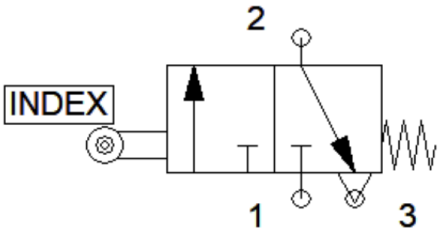
In some cases, a value may be described as “normally open” or “normally closed”:

3/2-way directional control valve, normally closed Note: The path to output 2 is blocked for default position, hence referred to as “normally closed”.	
3/2-way directional control valve, normally open Note: The path to output 2 is blocked for default position, hence referred to as “normally open”.	

Typically, Port 1 is for compressed air input, 2 and 4 for outlets to other components, and 3 and 5 to reject compressed air, or exhaust ports.

One actuation method to take note of are roller-lever actuated valves, where it has a roller limit switch at the end. This is most useful in detecting when a piston has fully extended, by placing it at the start or end of the piston.



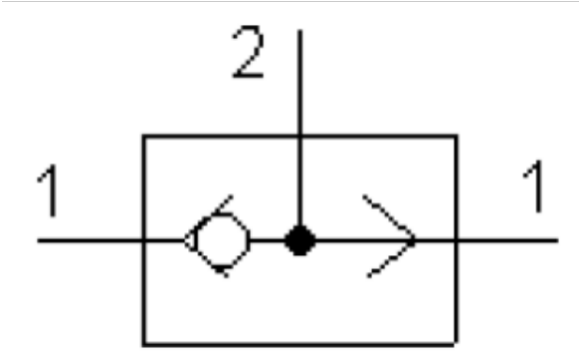


We represent it like so, with a box having a label representing a position along a certain piston.

There are also other valves that are not categorised as directional valves:

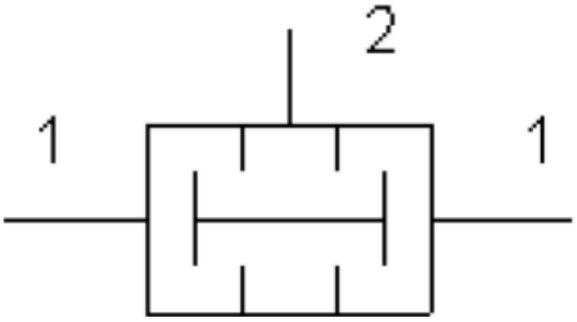
Shuttle Valve

Acts as an OR gate, represented like so:



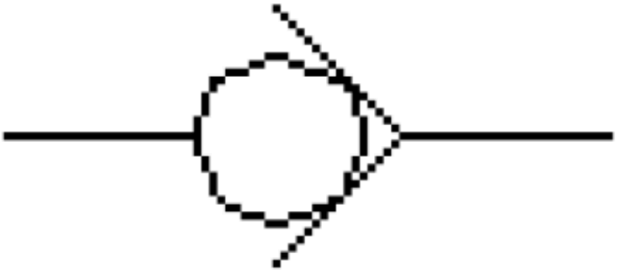
Two Pressure Valve

Acts as an AND gate, represented like so:



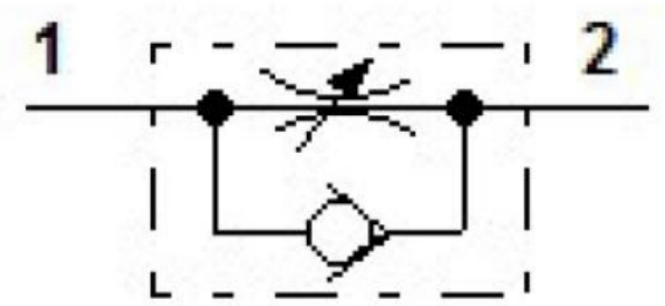
Check Valve

A one way valve, only allows flow in the direction that pushes the ball out:



One-Way Flow Control Valve

Regulates flow in one direction but allows free flow in the other, a combination of a nozzle and a check valve in parallel:



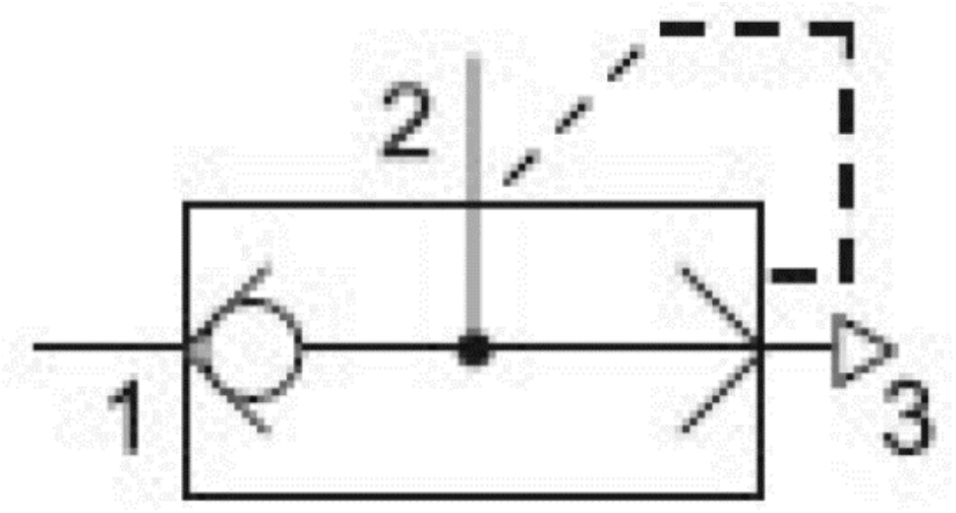
In this case, air is limited as it flows from 1 to 2 through the adjustable nozzle, but free flow is allowed from 2 to 1 through the check valve.

In the case of double acting cylinders, we usually place this so that it limits the airflow out, so for a forward stroke the one way flow check valve would be placed so that it limits air flowing out of the cylinder. This allows the pressure doing work to be unaffected and still supply the same amount of pressure as usual, maximising the effective force produced.

ME!!!!!!!!!!!!!!!!!!!!

Quick Exhaust Valves

The quick exhaust value has a large orifice to allow quick rejection of air.



When air flows from 1 to 2, port 3 is blocked and it acts as a normal valve.

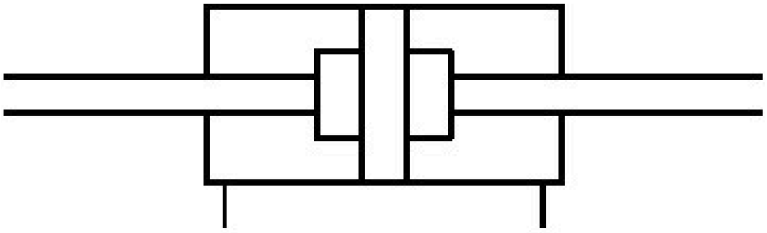
However, when air flows from 2, port 3 acts as a large exhaust and quickly rejects air out, which can be used to speed up double acting

cylinders by quickly exhausting the air out from the non-working section.

Other Pneumatic Actuators (NOT TESTED??????????????)

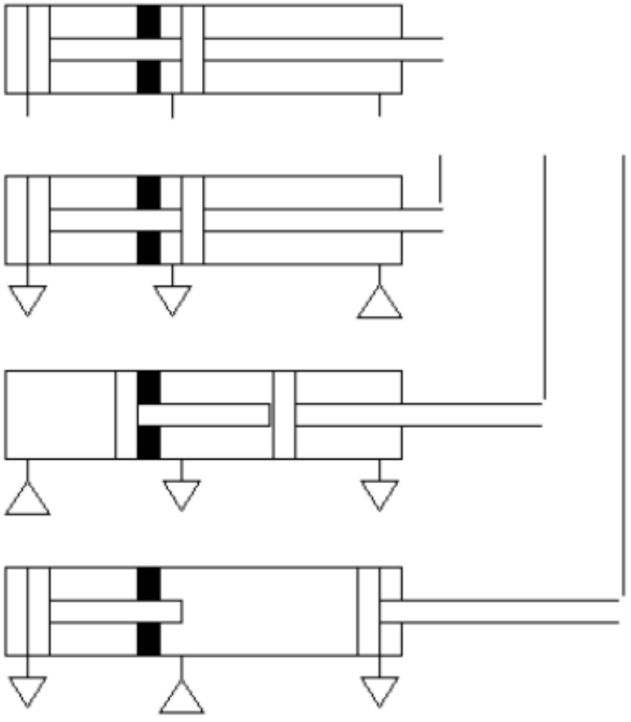
Through Rod Cylinder

The piston rod extends on both sides, giving a more rigid construction and better stability. Equal force is produced due to the same acting area.



Multi-Position Cylinder

By joining two double acting cylinders in series, one cylinder can push the other piston to have the piston reach 3 possible states.



Rodless Cylinder

No piston rod is present and instead any components are coupled onto the sliding piston, either mechanically or magnetically.

Rotary Vane Actuator

Rack and Pinion Rotary Actuator

Vane Air Motor

Pneumatic Gripper

Vacuum Gripper

Sequential Pneumatic Circuits & Cascading Method

i dont think i need notes for this